

COMP527 - CA Assignment 2 - Preprocessing

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1 About the Data

The data has 1760 sentences from various texts. A quick read of some of them shows that it's snippets from Shakespeare's *Coriolanus*, a New York noir novel, food-related writing and other miscellaneous or nonsensical phrases.

2 Loading the Data

The data was loaded as a Pandas data frame for easier clean-up and manipulation.

3 Cleaning the Data

The data will be vectorised with TF-IDF. For this to work properly, it is necessary to homogenise the text. Capitalisation needs to be removed, so words like 'sword', and 'Sword' are treated equally. Also, since this algorithm treats each sentence as a vector and each word as a component of said vector, punctuation and special characters become meaningless and need to be removed thusly.

With regex formatting the `clean_sentence` function was defined to replace unwanted characters with whitespace. Then, a `clean_data` function was created to apply the cleaning line by line, and label any empty lines with the word "empty". Both functions are applied to the data frame. To test the code, when run directly, `preprocessing.py` creates a .csv file with the outcome.

4 TF-IDF Parameters

Two parameters were set to further filter the vocabulary before vectorisation. `min_df=2` discards any word appearing in fewer than two sentences, as such words are likely proper nouns, typos, or overly specific terms that would not generalise across clusters. `max_features=4500` caps the vocabulary at the 4500 most frequent terms, keeping the resulting matrix computationally manageable and reducing noise from uncommon words.

5 Further Steps

Normalisation and LSA needed to be applied to the vectorised data. This was done within the actual modelling code, and will thus be explained in its documentation.